

```
# Here we use onnxruntime backend.

# Prepare backend session with ONNX model
sess = backend.prepare(model_new, 'CPU')

# Execute on backend
output = sess.run(data)

# Output
print("Output:", output)
```

The above example uses the *tf2onnx* package, which is a converter for TensorFlow to ONNX. For many frameworks, these converters are natively available, and for others there is an exclusive tool for conversion. A complete list of converters available for various frameworks is shown in Figure 2.

ONNX Model Zoo

To enable developers to start using ONNX, the community has hosted Model Zoo (<https://github.com/onnx/models>), which is a collection of pretrained state-of-art models in the ONNX format for image classification, object detection, image segmentation, gesture analysis, image manipulation, speech and audio processing, machine translation, language modelling and the visual question answering dialogue. A wide variety of Jupyter notebooks is available under this project, which helps developers to start with ease.

Deployment

ONNX deployment is possible in various targets starting from Android and iOS devices to AWS Lambda, Amazon SageMaker and Azure back-ends. A range of tutorials demonstrating various deployment scenarios can be found at <https://github.com/onnx/tutorials#end-to-end-tutorials>.

ONNX Runtime

ONNX Runtime is a new initiative from Microsoft towards ONNX's very own deployment runtime environment for ONNX models. It supports CUDA, MLAS (Microsoft Linear Algebra Subprograms), MKL-DNN and MKL-ML for computation acceleration. There is ongoing collaboration

Framework / tool	Installation	Exporting to ONNX (frontend)	Importing ONNX models (backend)
Caffe	apple/coremltools and onnx/onnxmltools	Exporting	n/a
Caffe2	part of caffe2 package	Exporting	Importing
Chainer	chainer/onnx-chainer	Exporting	coming soon
Cognitive Toolkit (CNTK)	built-in	Exporting	Importing
Apple CoreML	onnx/onnx-coreml and onnx/onnxmltools	Exporting	Importing
Keras	onnx/keras-onnx	Exporting	n/a
LibSVM	onnx/onnxmltools	Exporting	n/a
LightGBM	onnx/onnxmltools	Exporting	n/a
MATLAB	onnx converter on matlab central file exchange	Exporting	Importing
Menoh	pfnet-research/menoh	n/a	Importing
ML.NET	built-in	Exporting	Importing
Apache MXNet	part of mxnet package docs github	Exporting	Importing
PyTorch	part of pytorch package	Exporting, Extending support	coming soon
SciKit-Learn	onnx/sklearn-onnx	Exporting	n/a
TensorFlow	onnx/onnx-tensorflow and onnx/tensorflow-onnx	Exporting - ONNX-Tensorflow Exporting - Tensorflow-ONNX	Importing (experimental)
TensorRT	onnx/onnx-tensorrt	n/a	Importing

Figure 2: ONNX converter support (<https://github.com/onnx/tutorials>)

to support Intel MKL-DNN, nGraph and NVIDIA TensorRT. ONNX Runtime supports Python, C#, C and C++ API on Windows, Linux and Mac operating systems.

ONNX Runtime is released as a Python package in two versions—*onnxruntime* is a CPU target release and *onnxruntime-gpu* has been released to support GPUs like NVIDIA CUDA. With hardware acceleration and dedicated runtime for ONNX graph representation, this runtime is a value addition to ONNX.

Contributors

ONNX is licensed under MIT. It is supported by a wide range of community members from across academic institutions like the Technical University of Munich and MIT, from Facebook and Microsoft, along with many freelancers and anonymous contributors. 

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